SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I Sunil Kumar I S (192324155)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Sunil

Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.

Electronic Codebook (ECB) mode encrypts each plaintext block **independently** using the same encryption key.

- If two plaintext blocks are **identical**, ECB produces **identical ciphertext blocks**.
- As a result, repeated patterns in the plaintext remain visible in the ciphertext.
- An attacker can identify these patterns and gain information about the original data, even without knowing the encryption key.
- This makes ECB **unsuitable for transmitting sensitive or repetitive data**, such as financial records.

Example:

- Plaintext: `Salary = $5000 | Salary = $5000`
- ECB Ciphertext: `X1 | X1` (same ciphertext block repeated)

B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

Justification:

- CBC uses an **Initialization Vector (IV)** and XORs each plaintext block with the previous ciphertext block before encryption.
- Even if two plaintext blocks are identical, their ciphertext blocks will be **different** because each block depends on the previous encrypted block.
- This removes visible patterns from the ciphertext and provides much better confidentiality than ECB.

Advantages of CBC:

- Hides repeating patterns in plaintext.
- More secure for transmitting financial and confidential data.
- Widely used for secure block encryption (when implemented with a random IV and proper integrity protection).

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.

Blowfish

B. Justify your choice based on security strength and efficiency.

- **High Security:** Blowfish uses a **64-bit block size** and supports **variable key lengths from 32 to 448 bits**, making it much more secure than DES and generally stronger than 3-DES in practical use.
- **Better Performance:** Blowfish is **faster than 3-DES** because it performs fewer encryption operations while maintaining strong security.
- **DES Limitation:** DES uses a **56-bit key**, which is vulnerable to brute-force attacks and is no longer considered secure.
  because it encrypts data three times, making it less efficient for banking transactions.
- **Suitable for Banking:** Blowfish provides a good balance of **strong security** and **moderate performance**, making it the best choice among the given options.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.

- Both users publicly agree on a **prime number (p)** and a **generator (g)**.
- User A chooses a **private key (a)** and computes the public key: $\mathbf{A = g^a \bmod p}$.
- User B chooses a **private key (b)** and computes the public key: $\mathbf{B = g^b \bmod p}$.
- They exchange their public keys over the public network.
- User A computes the shared secret: $\mathbf{K = B^a \bmod p}$.
- User B computes the shared secret: $\mathbf{K = A^b \bmod p}$.
- Both obtain the **same shared secret key**, which is then used for secure encryption.

B. Identify a possible security threat and suggest a solution to mitigate it.

Threat:

- **Man-in-the-Middle (MITM) Attack:** An attacker intercepts the exchanged public keys and establishes separate keys with both users, allowing the attacker to read or modify the communication.

Solution:

- **Authenticate the key exchange** using **Digital Signatures** or **Digital Certificates (PKI)** to verify the identity of both users.
- This prevents attackers from impersonating either party during the key exchange.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.

RSA for Encryption and Decryption

1. Each user generates an **RSA key pair**:
   - **Public Key –** Shared with others.
   - **Private Key –** Kept secret by the owner.
2. The sender encrypts the email using the **recipient's public key**.

3. The encrypted email is sent over the network.
4. The recipient decrypts the email using their **private key**.
5. Since only the recipient has the private key, only they can read the message.

B. Explain how key distribution and management play a role in maintaining security.

Key Distribution and Management

- **Public keys** must be distributed securely so users can trust they belong to the correct person.
- **Digital Certificates** issued by a **Certificate Authority (CA)** are used to verify the authenticity of public keys.
- **Private keys** must be stored securely and never shared.
- Regular **key renewal, revocation, and backup** help maintain long-term security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.

Constraints of IoT Devices

- **Limited processing power** – IoT devices have low-performance processors.
- **Limited memory and storage –** They cannot handle large cryptographic operations efficiently.
- **Low battery power –** Encryption should consume minimal energy.
- **Limited bandwidth –** Smaller keys and messages are preferred for faster communication.
- **Real-time communication –** Fast encryption and decryption are required.

B. Recommend a suitable cryptographic approach and justify your answer.

Recommendation and Justification

Recommended Approach: Elliptic Curve Cryptography (ECC)

Justification:

- ECC provides **strong security with much smaller key sizes** than RSA.
- Smaller keys require **less processing power, memory, and battery**, making ECC ideal for resource-constrained IoT devices.
- ECC offers **faster key generation and key exchange** with lower communication overhead.
- RSA requires much larger keys (e.g., **2048 bits**) to achieve comparable security, making it less efficient for low-power IoT devices.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand